Dhanush Mathivanan

Arlington, VA • 240-385-2816 • dhanush.mathivanan@gwu.edu • linkedin.com/in/dhanush-mathivanan-a65202120
github.com/DhanushGWU1995 • dhanushgwu1995.github.io/dhanush-mathivanan-portfolio • techwithdhanush.blogspot.com

Professional Summary

Data and cloud engineer with 9+ years of experience building backend systems, microservices, and data pipelines on AWS, mostly serverless and event-driven, for clients including Porsche, Educational Testing Service, and Roche. Finishing an M.S. in Data Science at The George Washington University (May 2027), with coursework and projects in machine learning, deep learning, NLP, and LLM applications using RAG, built in Python and PyTorch. AWS Certified Solutions Architect and Developer (Associate). Seeking Data Engineer, Data Scientist, Machine Learning Engineer, and Software Engineer roles.

Technical Skills

Programming Languages: Python, SQL, TypeScript, JavaScript, R, Linux
Data Engineering: ETL/ELT pipelines, data modeling, Apache Airflow, data ingestion, batch and real-time processing, event-driven architecture, data warehousing, data validation (JSON Schema, Joi, AJV), PySpark, Kafka, AWS EMR, Pandas, Polars, NumPy
Machine Learning & AI: Deep learning, Natural Language Processing (NLP), generative AI, LLMs, RAG, hybrid search, embeddings, computer vision, transfer learning, predictive modeling, feature engineering, statistical analysis, hypothesis testing, A/B testing; PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, Hugging Face, Amazon Bedrock
Cloud & DevOps: AWS (Lambda, S3, API Gateway, DynamoDB, SQS, SNS, EventBridge, EC2, EMR, IAM, Cognito, CloudFormation, CDK, Connect, Lex), Terraform, CDKTF, Docker, Kubernetes, Jenkins, GitHub CI/CD
Software Engineering: System design, object-oriented programming (OOP), Agile, REST APIs, microservices, distributed systems, data structures and algorithms, OAuth/JWT; Node.js, Express.js, FastAPI, React, Angular, Socket.IO
Databases: Amazon Redshift, DynamoDB, MySQL, MongoDB, SQLite, DuckDB, Redis
Analytics, Monitoring & Testing: Tableau, Matplotlib, Splunk, Dynatrace, CloudWatch, Mocha/Chai, Jasmine/Karma, Git

Education

The George Washington University, Washington, DC — Aug 2025 – May 2027
Master of Science in Data Science
Coursework: Big Data Analytics, Cloud Computing, Machine Learning, Deep Learning, NLP, Data Warehousing

Anna University, Tiruchirappalli, India — 2012 – 2016
Bachelor of Engineering in Computer Science
Coursework: Computer Programming, Software Development, Algorithm Design, Web Application Development

Professional Experience

The George Washington University, Washington, DC
Technical Support Assistant — Aug 2026 – Present

- Built a contact center for university IT support on Amazon Connect, Amazon Lex, and AWS Lambda that lets callers resolve common requests through self-service; it cut Tier 1 manual escalations by an estimated 42%.
- Set up an event-driven pipeline with Amazon EventBridge, S3, Amazon Bedrock, and Amazon Q that publishes support updates every 15 minutes and lets staff search internal documentation.
- Supported device administration for 10,000+ users and 5,000+ network endpoints on Cisco CUCM and RedSky, and moved logs into Splunk Cloud to speed up root-cause analysis of critical incidents.
- Led planning, testing, and cutover for 2N IP intercoms integrated with Cisco VoIP, working with facilities, security, and IT teams; each go-live finished without disruption.

Probability Solutions LLC, Virginia
Data Science Intern — Jun 2026 – Aug 2026

- Researched how binary search trees and other tree-based index structures perform in database systems at scale.
- Wrote Python prototypes to benchmark search, insert, update, and delete (CRUD) operations on large inputs, and used statistical analysis to compare the trade-offs between implementations.

Bosch Global Software Technologies, Client: MHP Porsche (Audi AG), Germany
Technical Project Specialist / Lead Engineer — Mar 2024 – Aug 2025

- Built and deployed a workforce planning platform for Porsche on AWS serverless microservices (Lambda, API Gateway, DynamoDB, Cognito), with infrastructure as code in CDKTF and CloudFormation.
- Wrote the data ingestion pipelines for enterprise HR files: uploads to S3 trigger Lambda functions that validate each record against a JSON schema (Joi) and process it in real time.
- Connected legacy HR systems to the platform through SNS/SQS queues and Lambda to keep data in sync.

Accenture, Client: Educational Testing Service (ETS), USA

Custom Software Engineering Team Lead Sep 2021 – Mar 2024

- Cut exam result processing time by 50% by redesigning the scoring pipeline as serverless AWS services (Lambda, DynamoDB, Cognito, SNS/SQS, CloudFormation) for a platform that handles millions of exam records a year.
- Designed how score events enter the system: each event is checked with AJV schema validation, then processed by SQS-triggered Lambda functions, which kept high-volume scoring runs stable.
- Wrote DynamoDB maintenance scripts in Python and Node.js and built reusable TypeScript service components shared across the platform's APIs.

HCL Technologies, Client: Roche Holding AG, Switzerland

Lead Engineer – Full Stack Cloud Development Nov 2020 – Sep 2021

- Built REST APIs and reporting services in Node.js, Express.js, Angular, and MySQL that made diagnostic reporting workflows 40% faster.
- Replaced manual lab reporting with automatically generated PDF and Excel reports built from streamed data.
- Secured APIs with OAuth/JWT and set up AWS S3 storage for exchanging medical data between clinical systems.

Object Frontier Software, Client: International Flavors & Fragrances (IFF), USA

Software Engineer – Full Stack Development Aug 2018 – Nov 2020

- Built an offline-first desktop app with Electron and an embedded Node.js server that syncs SQLite and MySQL data, so manufacturing and supply-chain staff could keep working without a connection.
- Implemented NgRx (Redux) state management in Angular to support undo/redo on complex transactions, and wrote Node.js microservices on MySQL with Sequelize ORM.
- Added structured logging with Winston and optimized database-backed services to make issues easier to trace.

V2Soft, Client: Fiat Chrysler Automobiles (FCA), USA

Software Engineer – HMI Software Development Oct 2016 – Aug 2018

- Built HMI dashboards for vehicle infotainment and diagnostics in Angular, HTML5, Bootstrap, and JavaScript.
- Wrote MongoDB batch jobs and Node.js APIs to ingest and analyze vehicle diagnostic and telemetry data.
- Added real-time Socket.IO messaging between backend services and dashboards for live vehicle monitoring.

Projects

- **Torque Diagnostics (Big Data, NLP):** A retrieval-augmented generation (RAG) assistant that answers technician questions from ABB drive manuals and cites the page it used. PDFs are ingested into AWS S3 and processed with Polars, DuckDB, and PyArrow; hybrid search combines BM25 with embeddings, and a Qwen LLM writes the answer. A DeBERTa confidence check withholds answers the manuals do not support. Evaluated with precision, recall, F1, and MRR; runs as a Streamlit chat app on GPU EC2.
- **Enterprise Customer Risk Intelligence System (NLP):** An NLP app that analyzes customer complaints from the CFPB Consumer Complaint Database and ranks which need attention first; deployed on Hugging Face Spaces.
- **Facial Pain Detection (Deep Learning, Computer Vision):** Trained PyTorch models to classify facial images into four pain levels, comparing a custom CNN with transfer learning on VGG-16, ResNet-50, and EfficientNet-B3; used Grad-CAM to show which regions drove each prediction.
- **Heart Disease Prediction (Machine Learning):** Cleaned and analyzed CDC BRFSS 2024 survey data in Python and R to predict coronary heart disease risk, and published a prediction web app.
- **Health Status Prediction (Data Mining):** Studied how income relates to self-reported health in CDC BRFSS data and served a CatBoost model through a web app.
- **AI Parts Support Chatbot (Generative AI):** A parts support chatbot that classifies user intent and extracts appliance model numbers, built with Angular, Node.js, Express, TypeScript, SQLite, and the OpenAI/DeepSeek API.

Certifications & Awards

- AWS Certified Solutions Architect – Associate (SAA-C03), Amazon Web Services, 2024
- AWS Certified Developer – Associate (DVA-C02), Amazon Web Services, 2023
- Phi Delta Gamma Fellowship, The George Washington University, 2026
- Global Leaders Fellowship, The George Washington University, 2025
- Client Value Creation Award and Best People Award, Accenture, 2022